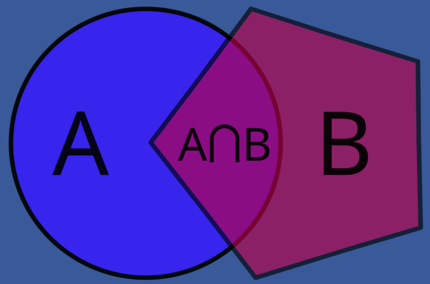
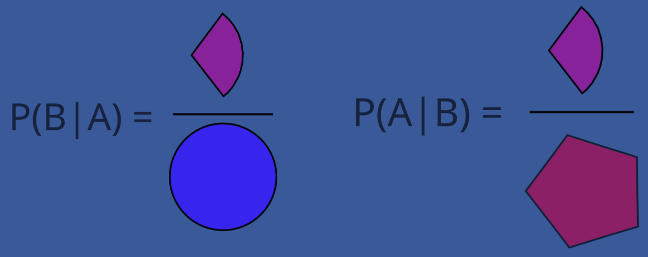
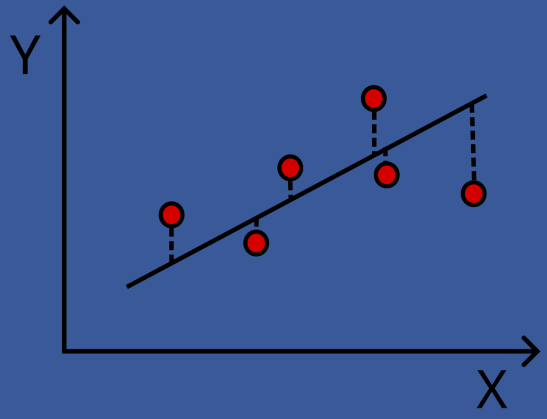
Maschinelles Lernen für die Chemie - Exercise 2







Bayes' Theorem, Data Types and Regression

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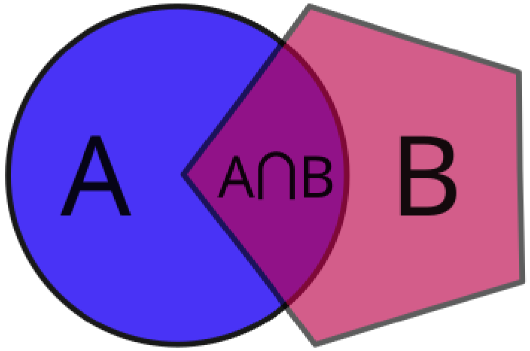
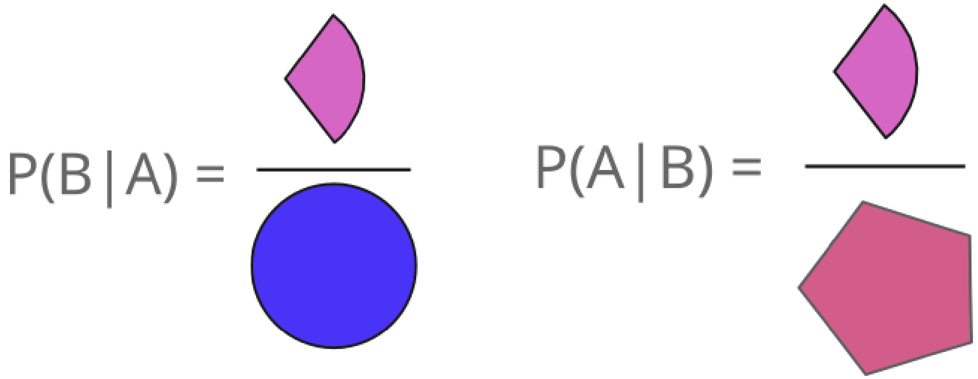
Probabilities and Notation

- $P(A)$: Probability of observing event A
- $P(B)$: Probability of observing event B
- $P(A|B)$: **Conditional** probability, the probability of event A occurring given that B is true



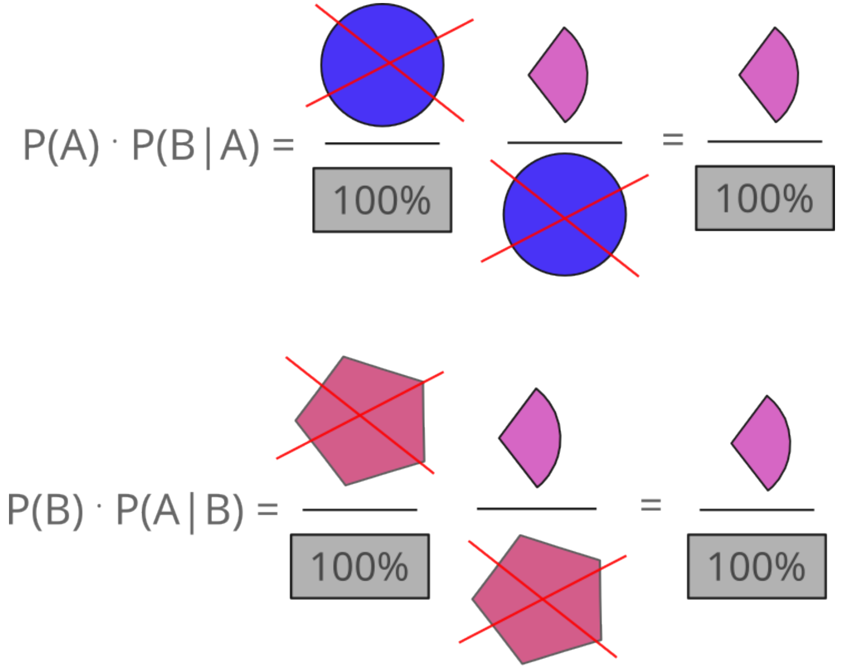


Bayes' Theorem

Mathematical rule for inverting conditional probabilities

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Proof:



$$= P(A) \cdot P(B|A)$$

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{P(B)}$$

Bayes' Theorem – Example

Testing for an infection (T: test positive, K infected people, G healthy people)

- Prevalence (people who get infected): $P(K) = 0.0002$
- Sensitivity (positive test if person is infected): $P(T|K) = 0.95$
- Specificity (positive test if person is healthy): $P(T|G) = 0.01$
- Amount of tested people: 100 000
- What is the probability that a random positively tested person is actually infected?

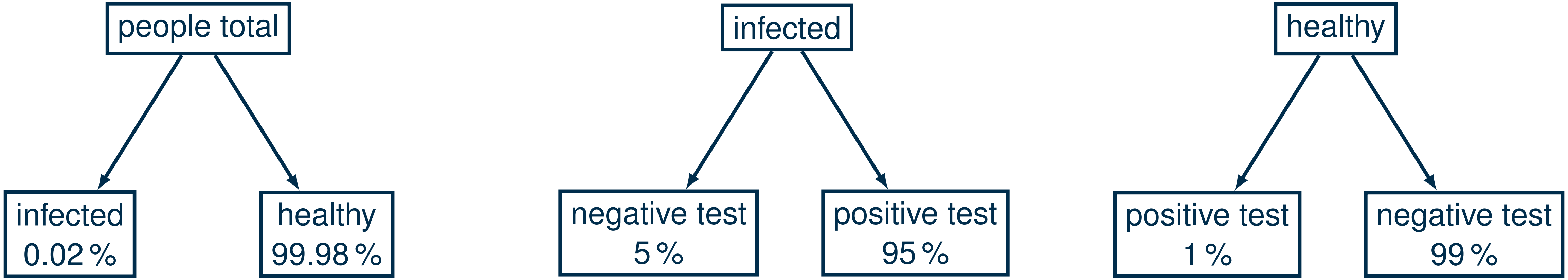
Bayes' Theorem – Example

Testing for an infection (T: test positive, K infected people, G healthy people)

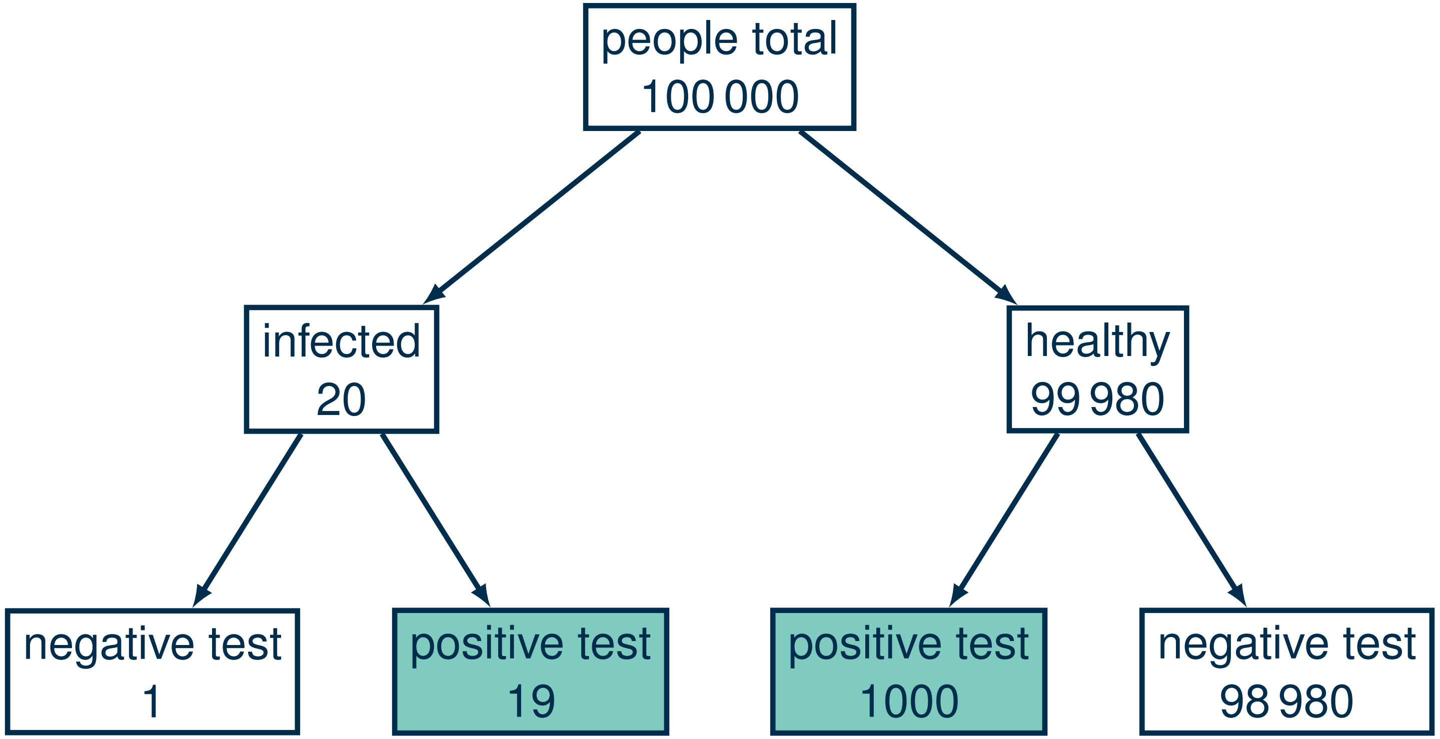
Prevalence $P(K) = 0.0002$

Sensitivity $P(T|K) = 0.95$

Specificity $P(T|G) = 0.01$



Bayes’ Theorem – Example



$$P(K|T) = \frac{P(T|K) \cdot P(K)}{P(T)}$$

Bayes' Theorem – Example

Solution:

$$P(K|T) = \frac{P(T|K) \cdot P(K)}{P(T)} = \frac{0.95 \cdot 0.0002}{0.95 \cdot 0.0002 + 0.01 \cdot 0.9998} \approx 0.0186$$

with the probability for the positively tested people

$$P(T) = P(T|K) \cdot P(K) + P(T|G) \cdot P(G)$$

What Are Data Types?

- Each variable has a data type
- Other programs: data type has to be set by the user
  Python: **dynamic typing**, data type is declared automatically
- Data type is important, because it determines:

1. the amount of memory used for this variable
2. the attributes of the variable (i. e. elements of a list can be accessed, accessing the elements of an integer doesn't make sense)

Examples for Data Types in Python

- Integer: `x = 2`
- Floats: `pi = 3.1415`
- String: `"Some random text"`
- List: `my_list = [1, 2, 3]`
- Dictionary: `my_dict = {key:  value}`
- Boolean: `condition = True`

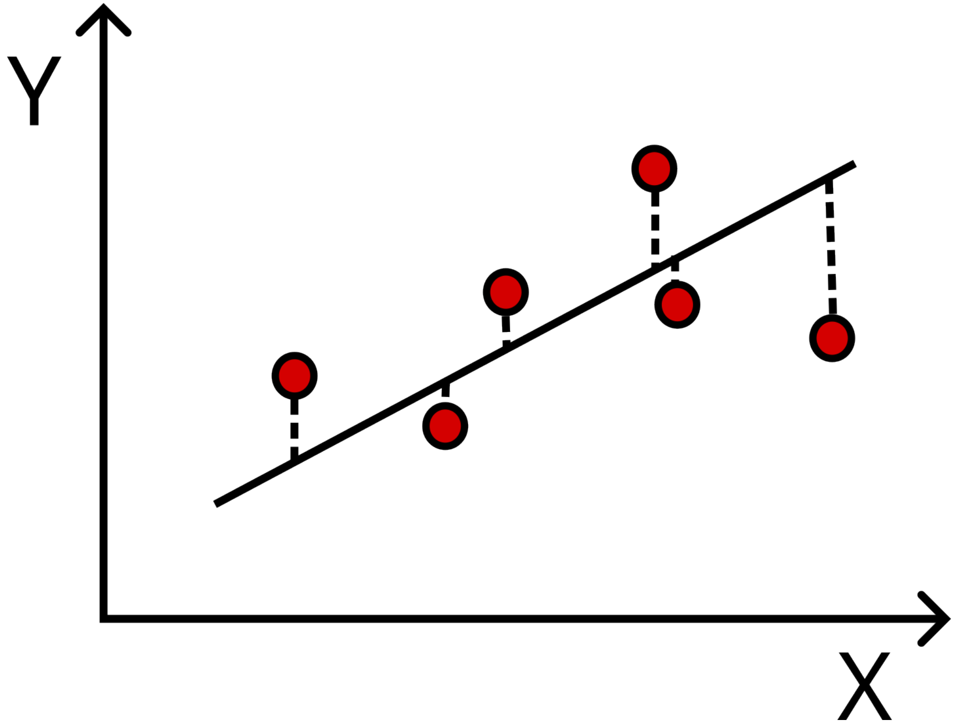
Type Hints

- Example: `x:  int = 2`
- States directly, that x should be an integer
- In the notebooks: used for hints what to do

```
x:  int
y:  np.ndarray
# Your code:
x = 100 # Sample size
y = np.arange(100) # Numpy array with all values from 0 to 99
```

Linear Regression

- Input vector (features) X
- Response (label, target) Y



$$f(X) = \beta_0 + \sum_j X_j \beta_j \tag{1}$$

With the set of training data (x_i, y_i), we minimize the **residual sum of squares**:

$$\mathrm{RSS}(\beta) = \sum_i (y_i - f(x_i))^2 \tag{2}$$

$$= \sum_i \left(y_i - \beta_0 - \sum_j x_{ij}\beta_j \right)^2 \tag{3}$$

Linear Regression – Matrix Notation

- Write the input vectors of the features X_j as a matrix $\boldsymbol{X}$ of shape $N \times p$ (N: number, p length of input vectors)
- Include intersect (bias) β_0 in the feature matrix $\boldsymbol{X}$ as a column of ones $\Rightarrow$ new shape: $N \times (p+1)$
- y: N-vector of outputs in the training set
- RSS can be expressed as:

$$\mathrm{RSS}(\beta) = (\boldsymbol{y} - \boldsymbol{X}\beta)^T(\boldsymbol{y} - \boldsymbol{X}\beta)$$

Linear Regression – Parameters

Minimizing the RSS in the matrix form leads to a unique solution for the parameters:

$$\hat{\beta} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$$

- $\boldsymbol{X}$: $N \times (p+1)$ matrix, N input vectors as rows, p length of input vectors, 1 in the first position (to account for β_0)
- $\boldsymbol{y}$: N-vector of outputs in the training set

Exercise:

- Features $\boldsymbol{X}$: medical information ($N_{\text{test}} \times 11$, first column consisting of ones)
- Target $\boldsymbol{y}$: diabetes or not (N_{test})

Linear Regression – Exercise

- Split the data in two random subsets for training and testing
  $\Rightarrow$ Scikit-learn: `train_test_split(feature_array, target_array, test_size=?, random_state=0)`

- Insert column of ones to the training set $\boldsymbol{X}$
  $\Rightarrow$ Numpy array of ones: `np.ones((train_set.shape[0], 1))`
  $\Rightarrow$ Prepend the array to the training set: `np.concatenate((ones_array, train_set), axis=1)`

- Calculate $\hat{\beta}$ with the prepended training matrix and the targets:
  Matrix multiplication: $\Rightarrow$ Numpy `np.matmul` or `@`
  Invert matrix: $\Rightarrow$ `np.linalg.inv`
  Transpose matrix: $\Rightarrow$ `matrix.T`

Linear Regression – Exercise

Or completely with Scikit-learn:

- Create a linear regression instance:
  $\Rightarrow$ `model: LinearRegression = LinearRegression()`

- Train your model:
  $\Rightarrow$ `model.fit(train_set, target_set)`

- Test the model by predicting the targets with the test set:
  $\Rightarrow$ `model.predict(test_set)`

- Save the parameters $\hat{\beta}$ as one Numpy array by inserting the interception at position 0 to the other coefficients:
  $\Rightarrow$ `np.insert(model.coef_, 0, model.intercept_, axis=0)`

Today's Notebooks

`2.1_Numpy_Crashcourse.ipynb`

- Optional, from Ex. 1

`1.1_LogisticRegressionTutorial.ipynb`

- Logistic regression tutorial to predict the subspecies of an iris flower
- Look here, if you need hints for the exercise notebook

`1.2_RegressionExercise.ipynb`

- Linear regression exercise as explained earlier
- Quantification of the results: refer to tutorial
- Polynomic regression: similar to linear regression, use appropriate Scikit-learn object
- Lasso regression as regularization example, again, different Scikit-learn object but otherwise similar

Literature

- Linear Regression

[1] Trevor Hastie, Tibshirani Robert, and Jerome Friedman. The Elements of Statistical Learning. 2nd ed. Springer Series in Statistics. NY: Springer New York, 2009. ISBN: 978-0-387-84858-7. DOI: https://doi.org/10.1007/978-0-387-84858-7.

- Bayes' theorem and probabilities

[2] Wikipedia contributors. Bayes' theorem. 2025. URL: https://en.wikipedia.org/wiki/Bayes%27_theorem (visited on 05/04/2025).

[3] Wikipedia contributors. Satz von Bayes. 2025. URL: https://de.wikipedia.org/wiki/Satz_von_Bayes (visited on 05/04/2025).